

**A  
Project Report  
On  
LiterSage: Literature Mastered**

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**Submitted at**



**DEPARTMENT OF INFORMATION TECHNOLOGY**  
**Faculty of Technology & Engineering, CHARUSAT**  
**Devang Patel Institute of Advance Technology and Research (DEPSTAR)**  
**At: Changa, Dist: Anand – 388421**  
**April 2026**

**CERTIFICATE**

This is to certify that the report entitled “**LiterSage: Litearture Mastered**” is a bonafied work carried out by **Patel Prince Pankajbhai(23DIT049)** under the guidance and supervision of **Prof. Chintal Raval** for the subject **IT368 – PROJECT - III** of 6<sup>th</sup> Semester of Bachelor of Technology in **Information Technology** at Devang Patel Institute of Advance Technology and Research (DEPSTAR), Faculty of Technology & Engineering (FTE) – CHARUSAT, Gujarat.

To the best of my knowledge and belief, this work embodies the work of candidate himself, has duly been completed, and fulfills the requirement of the ordinance relating to the B.Tech. Degree of the University and is up to the standard in respect of content, presentation and language for being referred to the examiner.

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**CERTIFICATE**

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## **DECLARATION BY THE CANDIDATES**

We hereby declare that the project report entitled “**LiterSage: Literature Mastered**” submitted by us to Devang Patel Institute of Advance Technology and Research (DEPSTAR), Changa in partial fulfilment of the requirements for the award of the degree of **B.Tech Information Technology**, from the Information Technology, DEPSTAR, FTE is a record of bonafide IT368 Project-III carried out by us under the guidance of **Prof. Chintal Raval**. We further declare that the work carried out and documented in this project report has not been submitted anywhere else either in part or in full and it is the original work, for the award of any other degree or diploma in this institute or any other institute or university.

(Patel Prince Pankajbhai – 23DIT049)

(Patil Bhagyesh Dinesh – 23DIT058)

(Prajapati Meet Girishbhai – 23DIT059)

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

## TABLE OF CONTENTS

|                                               |    |
|-----------------------------------------------|----|
| ABSTRACT .....                                | 1  |
| CHAPTER 1: INTRODUCTION .....                 | 2  |
| 1.1 BACKGROUND OF THE PROJECT.....            | 2  |
| 1.2 PROBLEM DEFINITION.....                   | 2  |
| 1.3 MOTIVATION FOR THE PROJECT.....           | 2  |
| 1.4 OBJECTIVES AND SCOPE OF THE PROJECT ..... | 2  |
| CHAPTER 2: LITERATURE REVIEW .....            | 3  |
| 2.1 EXISTING SOLUTIONS .....                  | 3  |
| 2.2 COMPARATIVE ANALYSIS.....                 | 3  |
| 2.3 HOW ChAI MATE DIFFERS .....               | 3  |
| CHAPTER 3: SYSTEM ANALYSIS.....               | 4  |
| 3.1 FUNCTIONAL REQUIREMENTS.....              | 4  |
| 3.2 NON-FUNCTIONAL REQUIREMENTS .....         | 4  |
| CHAPTER 4: TECHNOLOGY STACK.....              | 5  |
| CHAPTER 5: SYSTEM DESIGN.....                 | 6  |
| 5.1 USE CASE DIAGRAM.....                     | 6  |
| 5.2 ARCHITECTURE DIAGRAM.....                 | 6  |
| 5.3 MODULES/COMPONENTS .....                  | 7  |
| CHAPTER 6: TESTING.....                       | 8  |
| 3.1 TYPES OF TESTING.....                     | 8  |
| 3.2 TEST CASES AND RESULTS .....              | 8  |
| CHAPTER 7: RESULTS.....                       | 9  |
| CHAPTER 8: CHALLENGES FACED.....              | 13 |
| CHAPTER 9: CONCLUSION AND FUTURE SCOPE.....   | 14 |
| 3.1 CONCLUSION.....                           | 14 |
| 3.2 FUTURE SCOPE.....                         | 14 |
| REFERENCES.....                               | 15 |

## LIST OF TABLES

|                                       |   |
|---------------------------------------|---|
| TABLE 2.1 COMPARATIVE ANALYSIS .....  | 3 |
| TABLE 4.1 TECHNOLOGY STACK .....      | 5 |
| TABLE 6.1 TEST CASES AND RESULTS..... | 9 |

## LIST OF FIGURES

|                                                |    |
|------------------------------------------------|----|
| FIG 5.1 USE CASE DIAGRAM .....                 | 6  |
| FIG 5.2 ARCHITECTURE DIAGRAM.....              | 7  |
| FIG 7.1 LANDING PAGE .....                     | 10 |
| FIG 7.2 GUEST USER DOCUMENT UPLOAD PAGE.....   | 10 |
| FIG 7.3 LOADING PAGE .....                     | 11 |
| FIG 7.4 GUEST USER DASHBOARD .....             | 11 |
| FIG 7.5 MEMBER USER DOCUMENT UPLOAD PAGE ..... | 12 |
| FIG 7.6 MEMBER USER DASHBOARD.....             | 12 |
| FIG 7.7 MEMBER USER UPLOAD HISTORY PAGE.....   | 13 |
| FIG 7.8 MEMBER USER PROFILE PAGE.....          | 13 |

## **ABSTRACT**

LiterSage is an AI-powered document intelligence platform designed to transform the traditional reading and research experience into an interactive, highly efficient process. The project aims to enhance comprehension and productivity by providing automated, deep insights into uploaded documents for both guest and member users. Key features include comprehensive Document Summarization, Key Point Extraction, Pain Point Identification, and automated Title Generation, leveraging a combination of fine-tuned local machine learning models and the Groq API for high-performance natural language processing. Built with React, Tailwind CSS, and Framer Motion for a premium, futuristic frontend, and Node.js with Express for robust backend integration, LiterSage ensures a seamless and scalable solution. Registered members require accounts to access an exclusive, tabbed dashboard for deeper analysis, enhancing personalization and capabilities, while guest users can access limited core features. This innovative platform streamlines literature review, improves reading efficiency, and modernizes complex document analysis.



## CHAPTER 1: INTRODUCTION

### 1.1: Background of the Project

In recent years, the volume of digital literature, research papers, and complex documentation has grown exponentially, yet the traditional methods for reading and analyzing these texts remain largely manual and time-consuming. This leads to significant inefficiencies, such as information overload, difficulty extracting key insights, and a disconnected research experience for students, professionals, and avid readers. With the rapid advancement of artificial intelligence and natural language processing, users now expect intelligent, automated, and highly efficient tools to rapidly digest, summarize, and understand extensive written content.

### 1.2: Problem Definition:

Many students, researchers, and professionals face challenges processing large volumes of complex literature, often struggling to manually extract key insights, summarize core concepts, or identify critical pain points in a timely manner. Users frequently encounter difficulties quickly navigating and comprehending dense documents, while traditional reading methods and existing tools lack the advanced, automated analytical capabilities required to efficiently streamline the research and learning process.

### 1.3: Motivation for the Project:

The motivation behind LiterSage: Literature Mastered is to bridge the gap between readers and dense, complex documents by providing an interactive, AI-powered document intelligence platform. The goal is to enhance the research and reading experience for students, professionals, and guest users by offering rapid, automated assistance for document summarization, key point extraction, pain point identification, and insightful content analysis.

### 1.4: Objectives and Scope of the Project:

- To provide an interactive document intelligence platform for seamless extraction of insights from complex texts.
- To implement AI-powered models for automated document summarization, key point extraction, pain point identification, and title generation.

## CHAPTER 2: LITERATURE REVIEW

### 2.1: Existing Solutions:

Currently there are no existing solutions similar to a dedicated document intelligence platform like LiterSage, but alternatives of it are standard PDF readers, manual annotation tools, and generic AI chatbots. However, they focus primarily on basic text viewing or general conversational queries and not on specialized, automated features like targeted pain point identification, key point extraction, and seamless document summarization.

### 2.2: Comparative Analysis:

| PlatForm            | Document Viewing | Auto - Summarization | Insight Extraction | Analytical Dashboard |
|---------------------|------------------|----------------------|--------------------|----------------------|
| PDF Readers         | ✓                | ✗                    | ✗                  | ✗                    |
| Manual Tools        | ✓                | ✗                    | Limited            | ✗                    |
| Generic AI Chatbots | Limited          | ✓                    | Limited            | ✗                    |
| LiterSage           | ✓                | ✓                    | ✓                  | ✓                    |

Table 2.1 Comparative Analysis

### 2.3: How ChAI Mate Differs:

Literature Mastered enhances research and reading efficiency by integrating advanced AI-powered models for automated document summarization and targeted insight extraction, tailored specifically for complex literature and dense texts. Unlike existing solutions, it focuses equally on seamless Document Viewing, Auto-Summarization, Insight Extraction (Key and Pain Points), and providing a Unified Analytical Dashboard, ultimately benefiting both casual guest users and dedicated registered

## CHAPTER 3: SYSTEM ANALYSIS

### 3.1: Functional Requirements

- User Registration and Authentication
- Document Upload and Processing
- Local Models-Based Summarization and Key Point Extraction
- AI-Based Document Summarization
- AI-Based Key Point Extraction
- AI-Based Pain Point Identification
- AI-Based Title Generation
- Tiered Access Dashboards (Guest & Member)

### 3.2: Non-Functional Requirements

- **Usability:** Intuitive, visually premium interface ensuring seamless interaction and easy navigation for both guest and member users.
- **Scalability:** Ability to efficiently handle multiple simultaneous users and process various document sizes without performance degradation.
- **Performance:** High-speed document processing and minimal load times for generating complex, AI-driven text insights.
- **Security:** Robust protection of uploaded user documents and secure authentication mechanisms for registered member accounts.
- **Reliability:** High uptime and consistent AI model performance to ensure uninterrupted access to the platform's analytical services.

## CHAPTER 4: TECHNOLOGY STACK

| Category | Tools / Technologies                                                            | Purpose                                                                                        |
|----------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Frontend | React, Tailwind CSS, Framer Motion                                              | Premium User Interface, Animations, and UX                                                     |
| Backend  | Node.js, Express, PyPDF2, python-docx, python-pptx                              | Server-Side Logic, API Routing, and Document Parsing                                           |
| AI/ML    | Python, PyTorch, Hugging Face (facebook/bart-large-cnn and google/flan-t5-base) | bart-large-cnn for Summarization & Key Points, flan-t5-base for Pain Points & Title Generation |
| APIs     | REST APIs, Groq API                                                             | Communication between frontend/backend and Cloud AI processing                                 |

Table 4.1 Technology Stack

### Reason for Selection:

- **React** with **Tailwind CSS** and **Framer Motion** enables fast, dynamic, and visually premium UI development.
- **Node.js** and **Express** provide a highly scalable backend framework for efficient document handling and API routing.
- Hugging Face models (**BART & FLAN-T5**) via **PyTorch** are ideal for state-of-the-art, highly accurate NLP tasks like document summarization and targeted insight extraction.

## CHAPTER 5: SYSTEM DESIGN

### 5.1: Use Case Diagram

#### Actors:

- Guest User - Unregistered user with access to limited core features
- Registered Member - Authenticated user with access to the full, advanced analytical dashboard
- System Admin - Manages users, monitors system health, and maintains NLP models
- AI Processing System - The backend NLP models and Groq API handling the document analysis

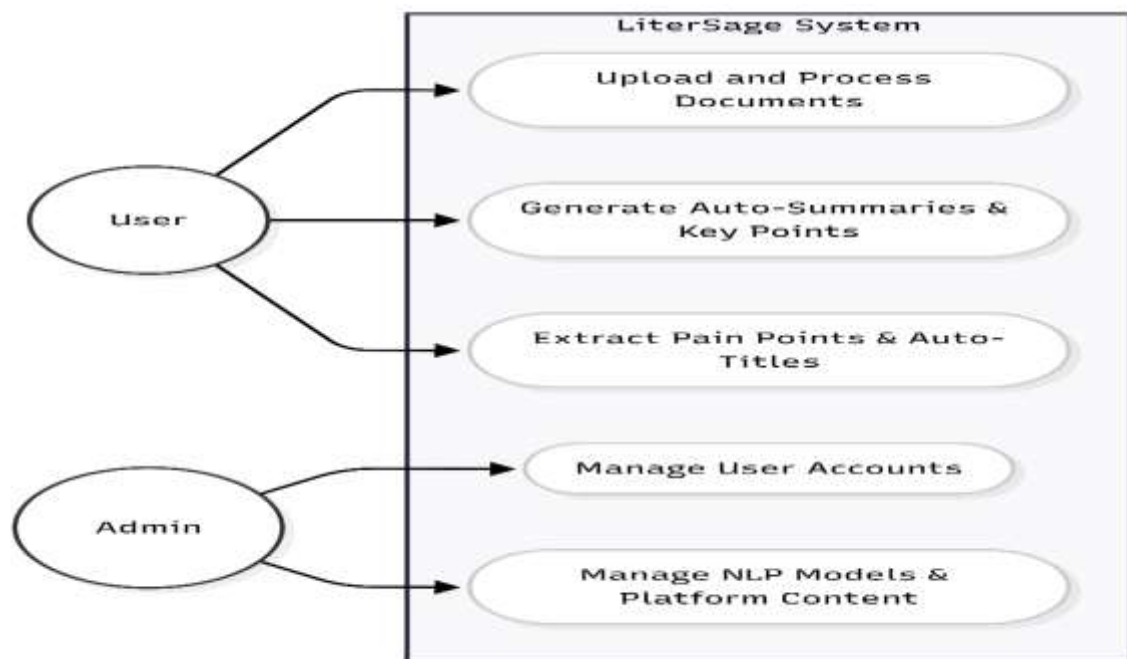


Fig 5.1 Use Case Diagram

### 5.2: Architecture Diagram: A three-tier architecture is used.

1. **Presentation Layer (Frontend)** – React + Tailwind CSS + Framer-Motion
2. **Application Layer (Backend)** – Node + Python
3. **Data Layer (Database)** – MongoDB Atlas

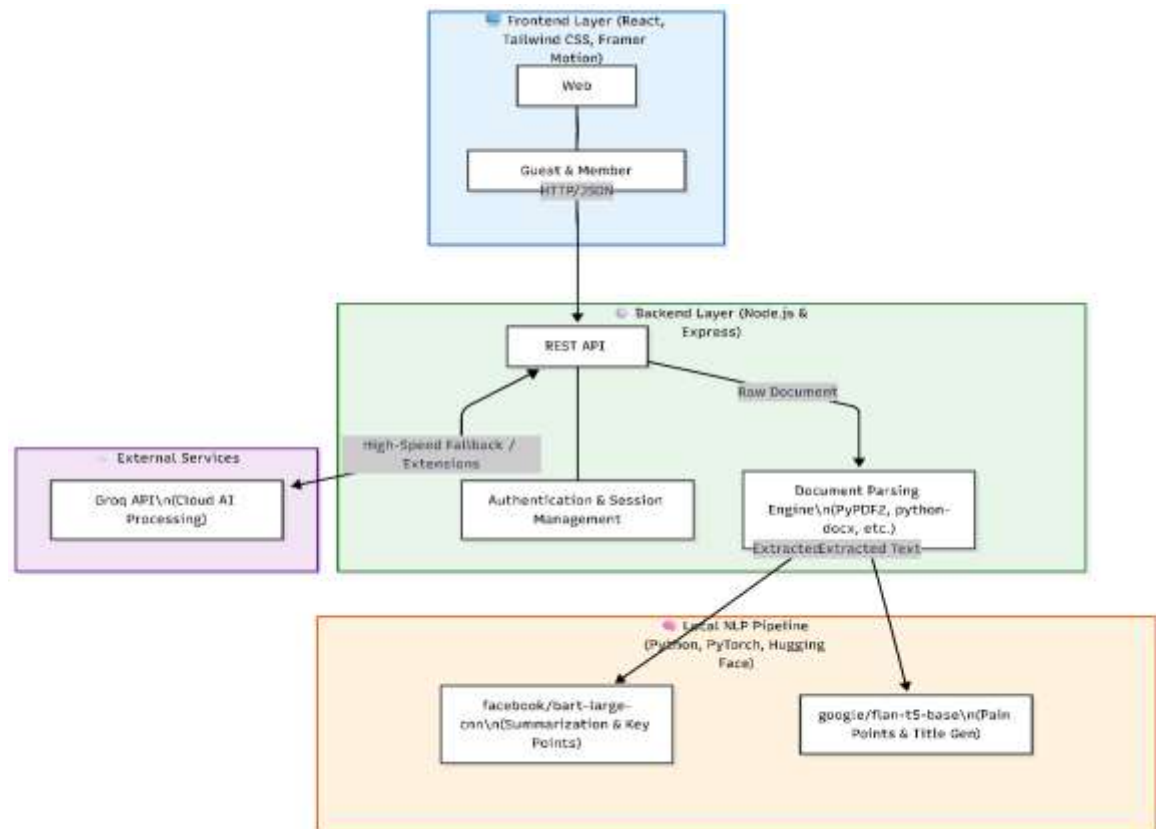


Fig 5.2 Architecture Diagram

### 5.3: Modules / Components

1. User Authentication
2. Document Upload & Parsing
3. AI Document Summarization
4. AI Key Point Extraction
5. AI Pain Point Identification
6. Automated Title Generation
7. Analytical Dashboard (Guest & Member)

## CHAPTER 6: TESTING

### 6.1: Types of Testing

- **Unit Testing** – Verified individual frontend components, API routes, and text parsing modules.
- **Integration Testing** – Tested the seamless flow between the React frontend, Node backend, and local models.
- **System Testing** – Checked the entire document intelligence pipeline functionality, from file upload to insight generation.
- **User Acceptance Testing (UAT)** – Verified platform usability and the accuracy of AI-generated insights with test users.

### 6.2: Test Cases and Results

| Test ID | Module               | Expected Result                                                                     | Actual Result     | Status |
|---------|----------------------|-------------------------------------------------------------------------------------|-------------------|--------|
| TC01    | User Authentication  | Successful login and session creation with valid credentials                        | Successful        | ✓      |
| TC02    | Tiered Dashboards    | Distinct UI features and access limits applied correctly for Guest vs. Member users | Works as expected | ✓      |
| TC03    | NLP Analysis Modules | Accurate extraction of summaries, key points, and titles from uploaded documents    | Working properly  | ✓      |

Table 6.1 Test Cases and Results

## CHAPTER 7: RESULTS

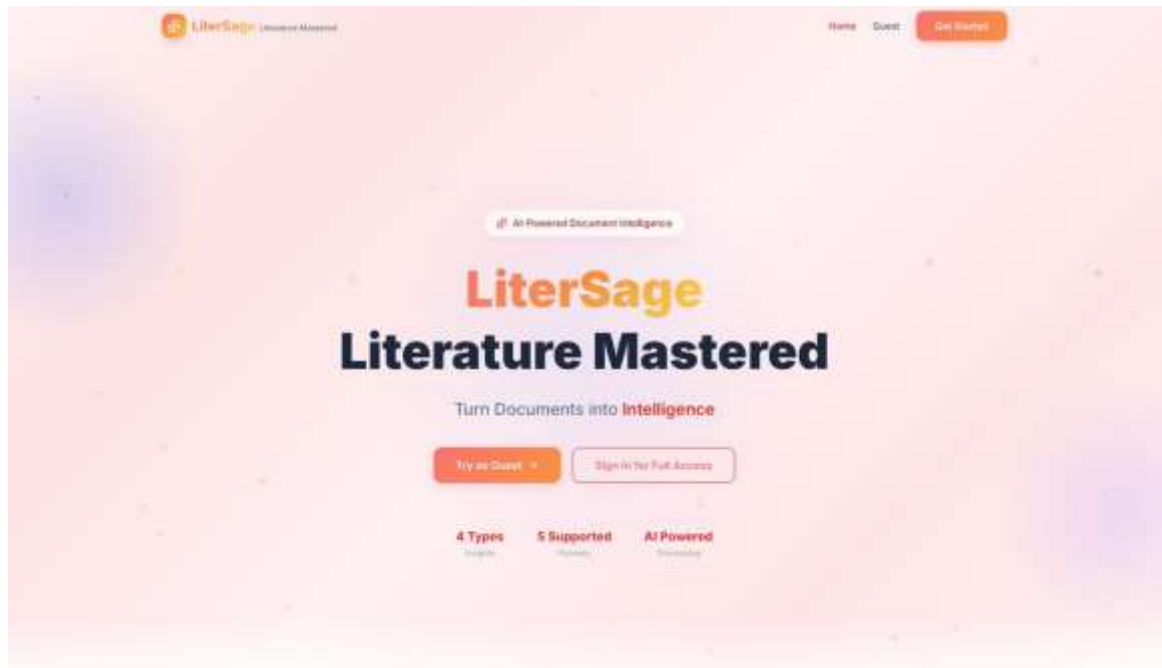


Fig 7.1 Landing Page

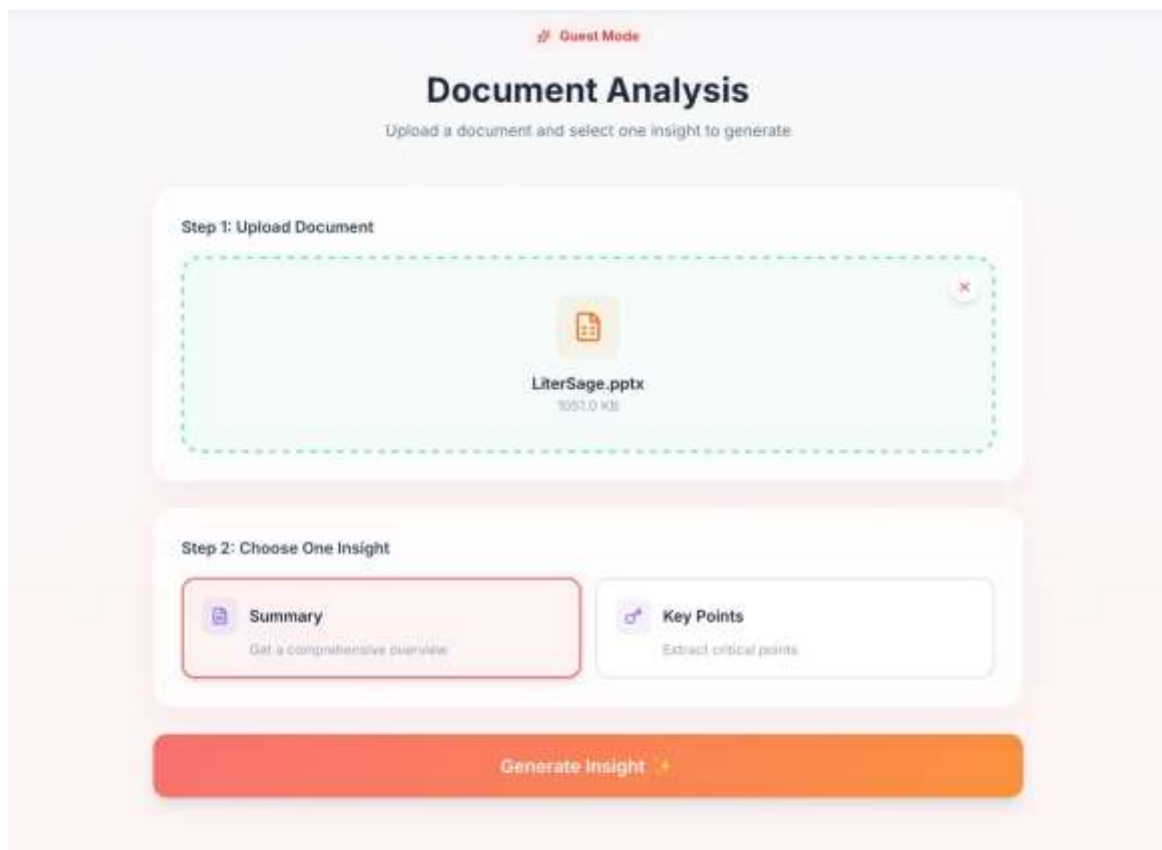


Fig 7.2 Guest User Document Upload Page





Fig 7.3 Loading Page

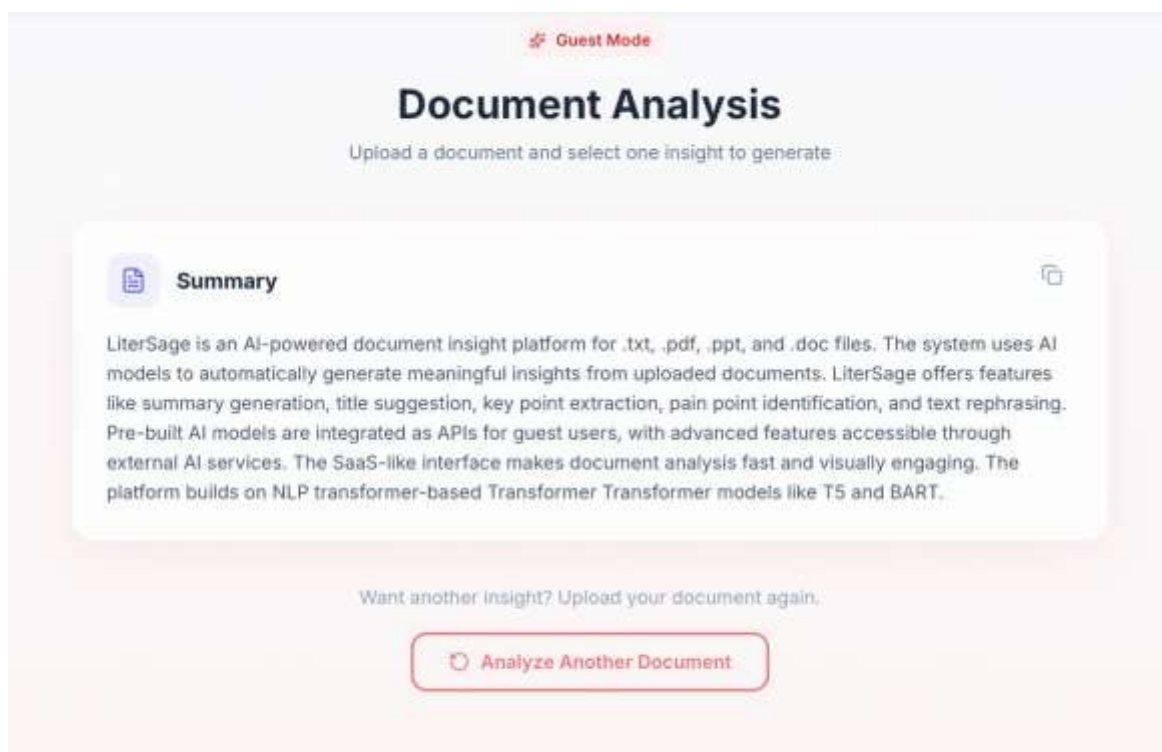


Fig 7.4 Guest User Dashboard

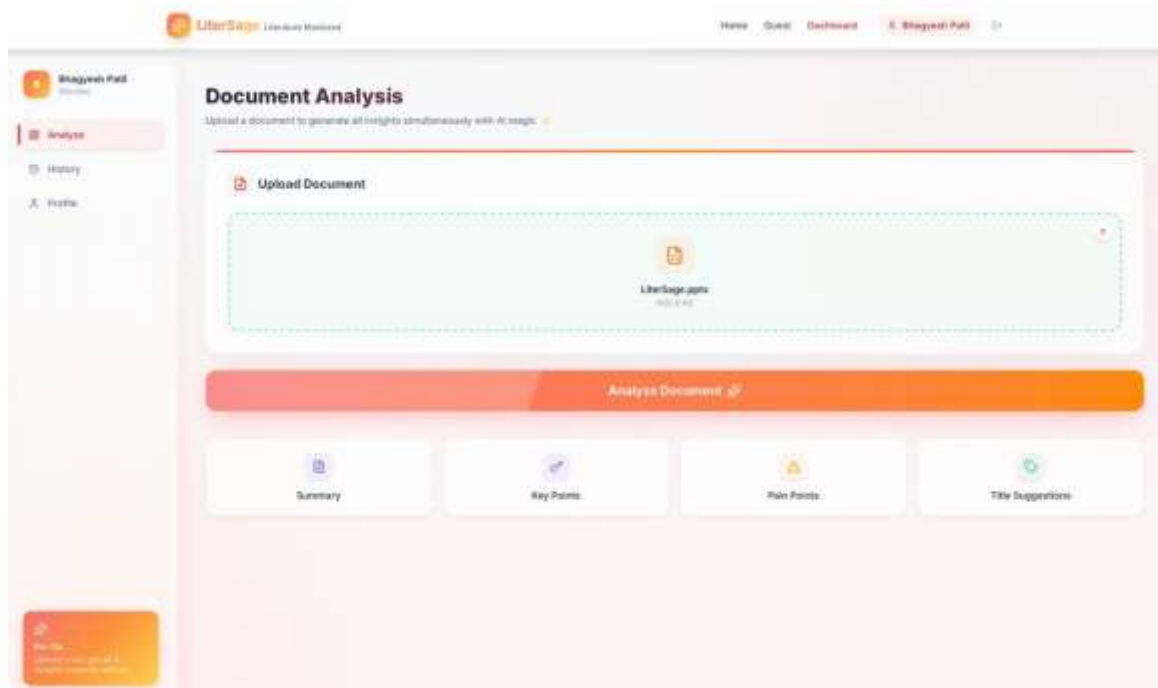


Fig 7.5 Member User Document Upload Page

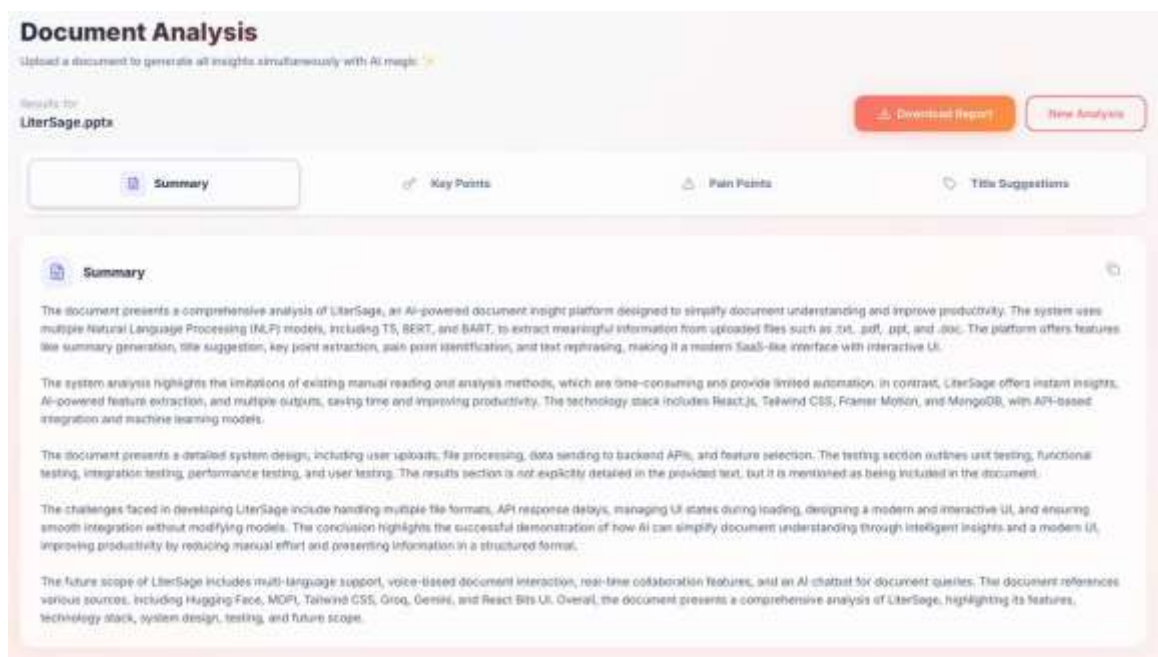


Fig 7.6 Member User Dashboard



Fig 7.7 Member User Upload History Page

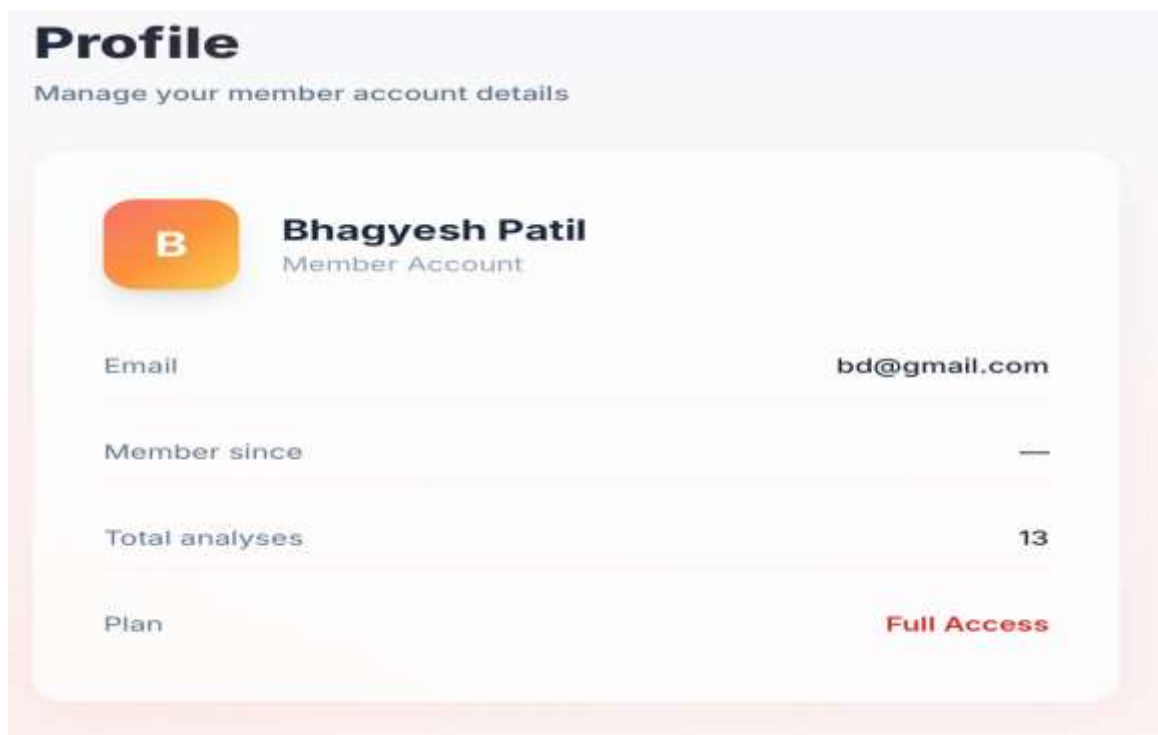


Fig 7.8 Member User Profile Page

## CHAPTER 8: CHALLENGES FACED

- **Fine-Tuning AI Models on Local Hardware:** Training and fine-tuning multiple NLP models (BART and FLAN-T5) locally on Apple Silicon required careful resource management. We encountered compatibility issues with PyTorch's Metal Performance Shaders (MPS), leading to crashes during unsupported operations, as well as strict memory constraints when orchestrating the unified training pipeline for four different models.
- **Managing Complex Document Parsing and Token Limits:** Processing unstructured, real-world documents (PDFs, DOCX) resulted in messy, unformatted text strings. Feeding this raw data into the local Hugging Face models was difficult because models like bart-large-cnn have strict maximum token limits, which long documents easily exceeded.
- **Resolving Environment and Port Conflicts:** During the integration of the Node.js backend and React frontend, we encountered severe environment issues, notably a port conflict where macOS AirPlay was occupying the default backend port (5000), preventing the API server from starting and disrupting client-server communication.
- **UI/UX Consistency and CSS Styling Conflicts:** Achieving the desired "premium, futuristic" aesthetic using Tailwind CSS and Framer Motion led to complex styling conflicts. Specifically, overriding default Tailwind utility classes caused unexpected layout shifts and rendering inconsistencies between the Guest and Member dashboards.

### Solutions:

- Configured PyTorch MPS CPU fallbacks and utilized PEFT to train models within local hardware constraints.
- Implemented dynamic text chunking to prevent documents from exceeding Hugging Face model token limits.
- Migrated the API server to port 5001 and updated the frontend proxy to bypass AirPlay port conflicts.

## **CHAPTER 9: CONCLUSION AND FUTURE SCOPE**

### **9.1: Conclusion**

LiterSage: Literature Mastered successfully achieves its goal of transforming the traditional, manual reading experience into an interactive, highly efficient AI-powered platform. It bridges the gap between users and complex literature, delivering rapid, automated extraction of document summaries, key points, pain points, and dynamic titles, all while ensuring secure, role-based tiered access and optimal research productivity.

### **9.2: Future Scope**

- Mobile App development for cross-platform accessibility.
- Integration of more advanced analytical modules, such as a 'Chat with Document' feature and sentiment analysis.
- Voice-based read-aloud feature for document summaries and key points.
- Multi-language support to process, translate, and summarize non-English literature.
- Integration with cloud storage platforms (Google Drive, Dropbox) and academic databases for direct document import.

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